

R Programming

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Assignment 3

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1. Create the following vectors

(a) $(0.1^3 0.2^1, 0.1^6 0.2^4, 0.1^9 0.2^7, \dots, 0.1^{36} 0.2^{34})$

(b) $(2, 2^2/2, 2^3/3, \dots, 2^{25}/25)$

(c) Execute the following lines which create two vectors of random integers which are chosen with replacement from the integers 0, 1, ..., 999. Both vectors have length 250.

```
set.seed(50)
```

```
xVec <- sample(0:999, 250, replace=T)
```

```
yVec <- sample(0:999, 250, replace=T)
```

Suppose $x = (x_1, x_2, \dots, x_n)$ denotes the vector xVec and $y = (y_1, y_2, \dots, y_n)$ denotes the vector yVec.

- Pick out the values in yVec which are > 600
- What are the index positions in yVec of the values which are > 600 ?
- What are the values in xVec which correspond to the values in yVec which are > 600 ?
- Sort the numbers in the vector xVec in the order of increasing values in yVec
- Pick out the elements in yVec at index positions 1, 4, 7, 10, 13, ...

2. (Webscraping)

- Read the csv dataset from the following webpage:

[R Assignment Webpage](#)